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Design and Development of Wireless LiFi Communication System using PAM

1. Introduction and Project Summary

This project presents the design and implementation of a basic Li-Fi (Light Fidelity) communication system using an analog input signal. Li-Fi is an optical wireless technology that transmits data through visible light, making it a potential alternative to traditional radio-based communication for short distances.

In this setup, Pulse Amplitude Modulation (PAM) is used as the core modulation technique. The complete system is built using discrete analog components on a breadboard, demonstrating the process of encoding and decoding signals through light. The main goal of this project is to transmit a pure sine wave wirelessly through a visible light link and successfully reconstructing it at the receiver end, focusing on the clarity and quality of the recovered waveform.

2. System Level Design and Working Principle

The overall system is divided into three main sections — Transmitter, Optical Channel, and Receiver/Demodulator.

Transmitter Section: PAM Signal Generation

The transmitter is responsible for generating the PAM-modulated signal and transmitting it through light. This is done using basic analog circuitry, without the use of operational amplifiers.

Input Signals (From Function Generator):

Message Signal: A 390 Hz sine wave acts as the baseband message.

Carrier Signal: A 4 kHz square wave used as the sampling pulse train.

- **PAM Formation:**
  Both the message and carrier signals are applied to a transistor-based switching circuit. The transistor acts as a sampling switch, allowing the carrier pulses to pick up the amplitude of the sine wave. The resulting pulse amplitudes vary in proportion to the instantaneous value of the message signal, forming the PAM waveform.

- **Optical Transmission:**
  The PAM signal then drives a **BJT transistor**, which controls the current through a **high-brightness LED**. As a result, the LED's light intensity varies according to the PAM signal, effectively transmitting the data through the visible light channel

Receiver Section: Optical Detection and Signal Recovery

At the receiver end, the transmitted optical signal is detected, converted back to an electrical form, and demodulated to retrieve the original message signal.

- **Optical-to-Electrical Conversion:**

A solar panel acts as the photodetector, converting the varying light intensity into a corresponding voltage signal.

- **Demodulation Process:**
  The detected voltage is passed through an RC low-pass filter to remove the high-frequency components (the 4 kHz carrier), thereby extracting the original low-frequency sine wave.
- **Output Verification:**
  The recovered signal is displayed on an oscilloscope and compared with the original input signal to evaluate the accuracy and fidelity of transmission.

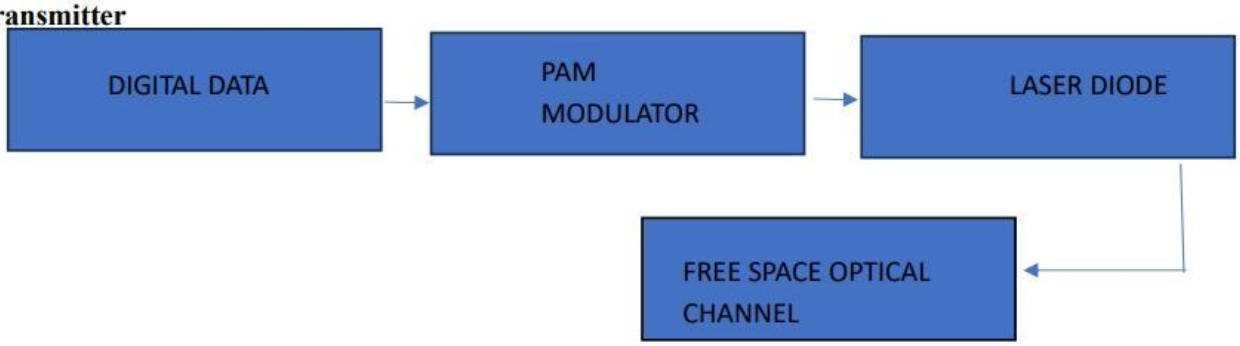
1 Key Components and Equipment

Component	Quantity	Role
Transistors (BJT)	2	PAM modulation
Solar Panel	1	Optical Receiver
High Brightness LED	1	Visible Light Transmitter
Resistors/Capactiors	1K,10K,10u,0.01u,100u,0.1uF,470u	Filtering and Biasing Components
Oscilloscope	1	Signal Analysis and Verification
Breadboard	2	For connecting the circuit
Connecting wires	As Required	For connecting the circuit
Transistors	1 (TIP41)	Preamplifier
Integrated circuits	3 (UA741CP, LM386)	Amplification

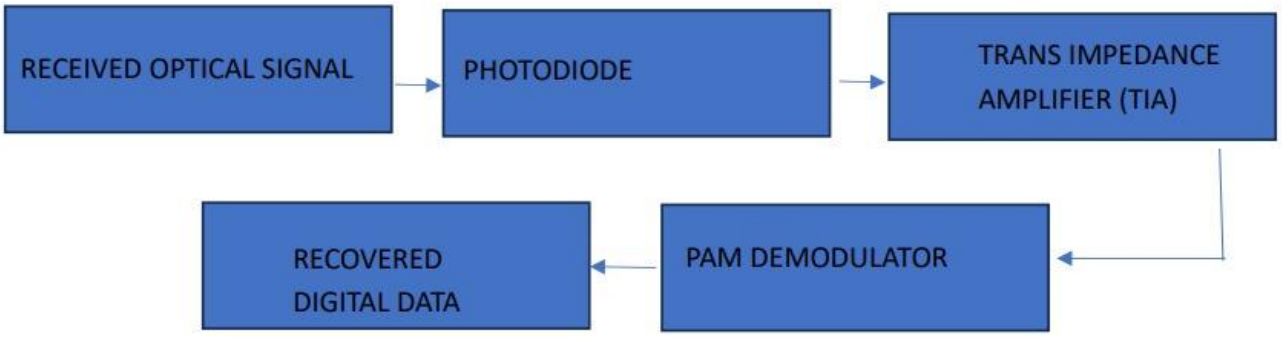
2 BLOCK DIAGRAM

BLOCK DIAGRAM:

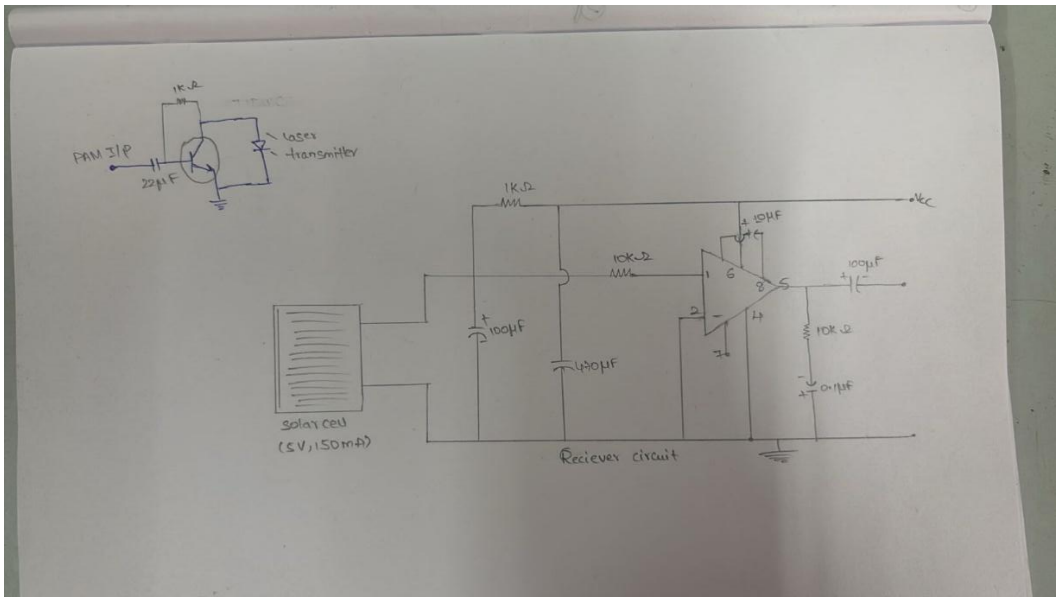
Transmitter



Receiver

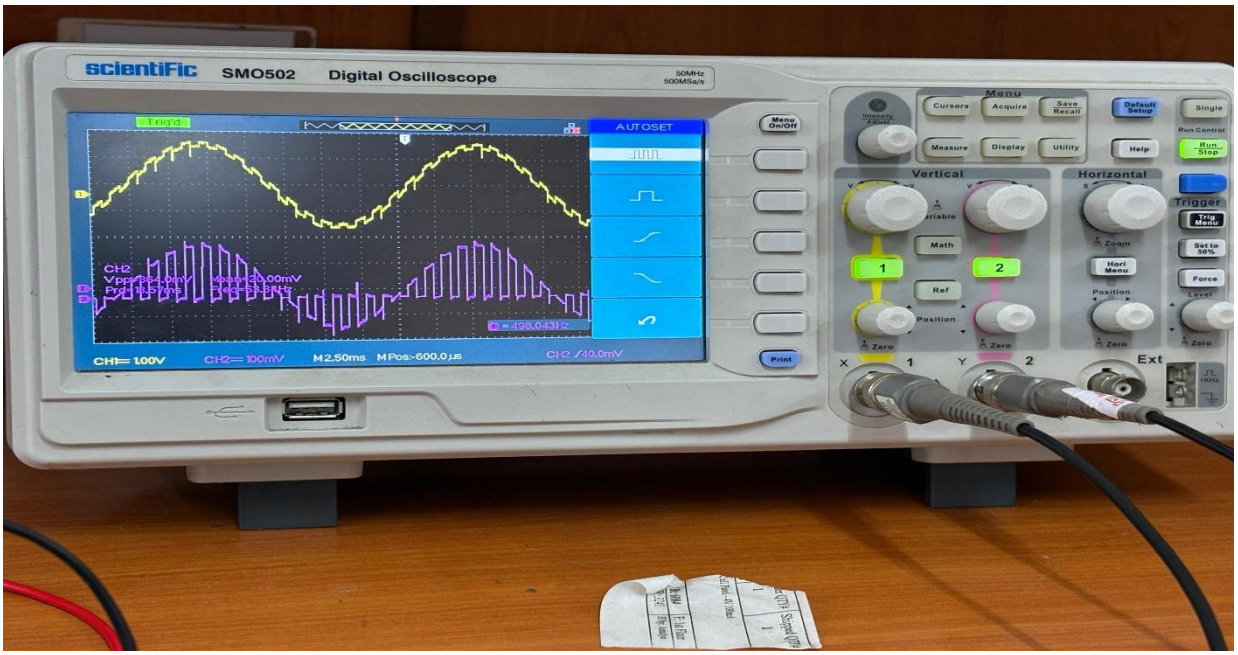


3 CIRCUIT DIAGRAM

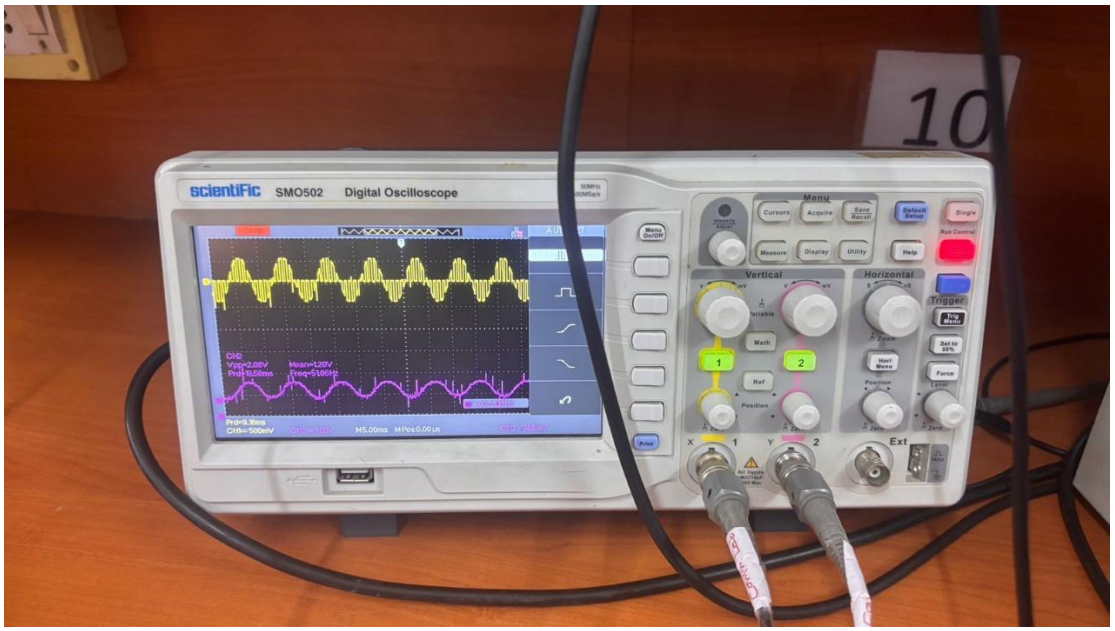


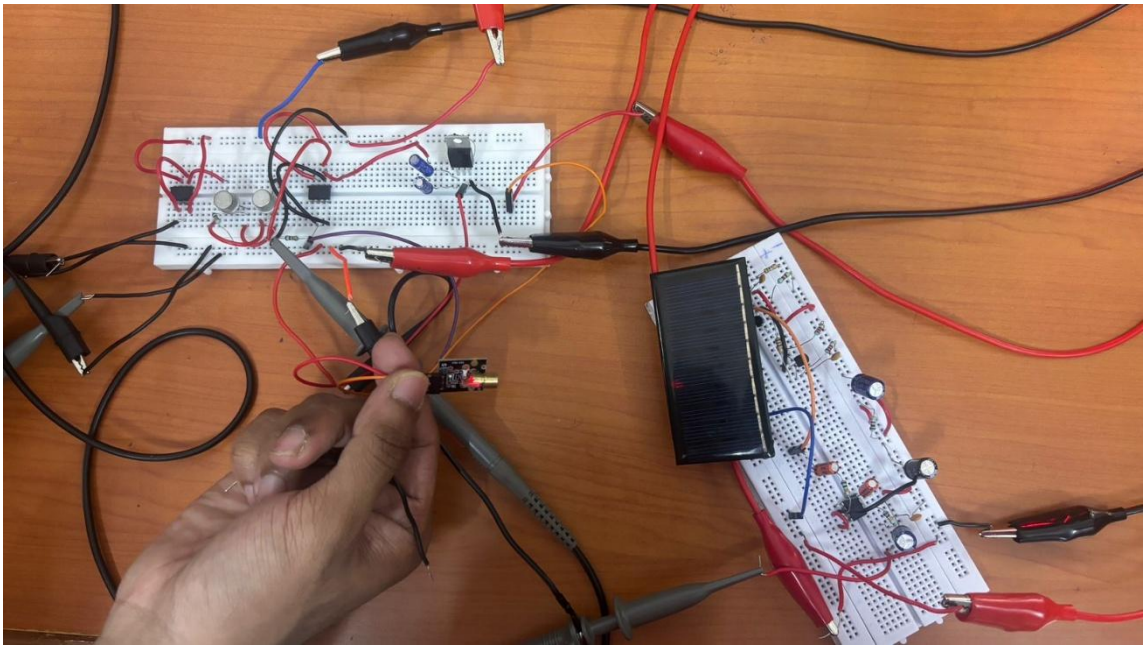
4 OUTPUT

- PAM Modulated Outpu



Received Signal From Solar Panel:





5 Results and Conclusion

The project successfully demonstrated the core principle of optical transmission but revealed a critical limitation in the fidelity of signal reconstruction.

- **System Parameters:** The message frequency was 390HZ sampled by a 4kHz carrier, which respects the Nyquist rate.
- **Final Outcome:** Despite extensive tuning and troubleshooting, the system was only able to recover the positive half-cycle of the 390Hz sine wave. The negative half-cycle was clipped to zero at the receiver.

Conclusion: This project successfully implemented a discrete-component Li-Fi link and validated the method of PAM transmission. However, the result confirms that transmitting a full AC analog signal via an LED (a uni-directional current device) is highly challenging.

6 Critical Challenges Faced and Technical Learning

Challenge	Technical Explanation	Solutions Attempted
Failure to Recover Negative Half-Cycle	The LED is a diode, which inherently allows current flow in only one direction. This clipped the negative portion of the sine wave modulation, resulting in the transmitted light intensity being zero during the negative half-cycle.	Multiple biasing techniques were attempted: Adding DC Offset, Adding External Voltage, Clamper Circuit implementation, and a Clipper Circuit for signal conditioning.
Learning Outcome	The inability to recover the complete output, even after multiple attempts, confirmed that overcoming the fundamental non-linearity of the LED optical source requires a robust, linear DC bias component to ensure the LED is always on (never drops to zero current) while still being modulated by the full swing of the sine wave.	

PART B

To compare the Bit Error Rate (BER) performance of NRZ and Manchester line coding schemes using a linear equalizer in an AWGN channel.

CODE

```
Start of document. Line 1 %% Compact: BER — NRZ vs Manchester over ISI channel with LMS EQ
Line 2 %% Compact: BER — NRZ vs Manchester over ISI channel with LMS EQ
%% Compact: BER — NRZ vs Manchester over ISI channel with LMS EQ
clc; clear; close all; rng(1);

% Params
N = 1e5; EbN0 = 0:2:14;
h = [0.9 0.5 0.2];          % ISI channel
h = h / norm(h);            % optional normalize
numTaps = 8; mu = 0.01;

% Data & encoders
bits = randi([0 1],1,N);
nrz = 2*bits-1;
man = 2*reshape([bits; 1-bits],1,[]) - 1;   % 2 samples/bit

% Channel
tx_nrz = conv(nrz, h, 'same');
tx_man = conv(man, h, 'same');

% Preallocate
BER_nrz = zeros(size(EbN0));
BER_man = zeros(size(EbN0));

for i = 1:length(EbN0)
    snr = EbN0(i);
    rx_nrz = awgn(tx_nrz, snr, 'measured');
    rx_man = awgn(tx_man, snr, 'measured');

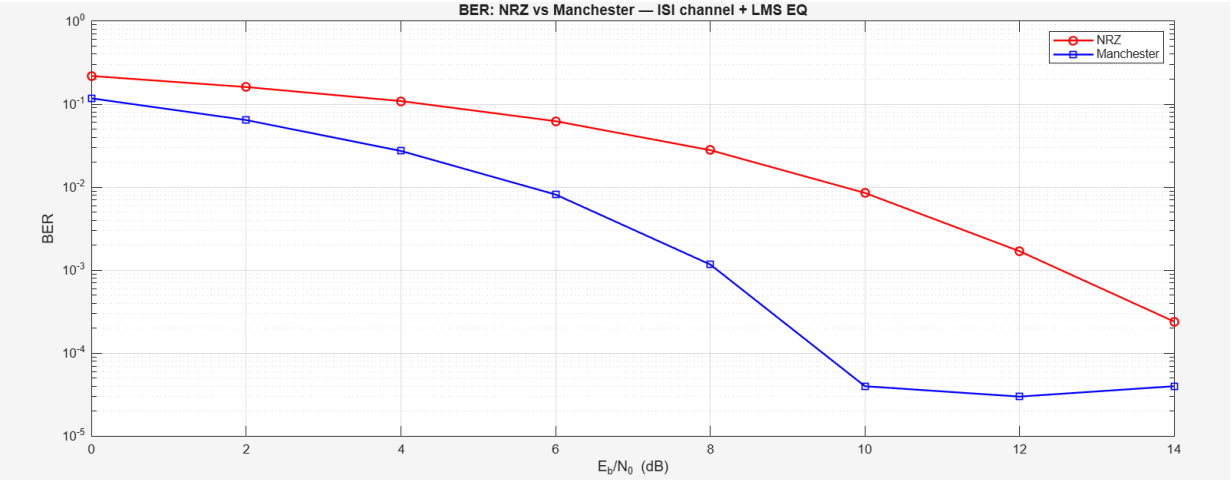
    % LMS equalizers (adapt rx -> original waveform)
    eqN = dsp.LMSFilter(numTaps,'StepSize',mu);
    [nrz_eq, ~] = eqN(rx_nrz.', nrz.');
    eqM = dsp.LMSFilter(numTaps,'StepSize',mu);
    [man_eq, ~] = eqM(rx_man.', man.');
```

```
    % Decisions
    dec_nrz = nrz_eq > 0;                          % column
    M = reshape(man_eq, 2, []);
    dec_man = M(1,:) > M(2,:);

    % BER
    BER_nrz(i) = mean(dec_nrz.' ~= bits);
    BER_man(i) = mean(dec_man ~= bits);
end

% Plot
figure; semilogy(EbN0, BER_nrz,'r-o',EbN0,BER_man,'b-s','LineWidth',1.3);
grid on; xlabel('E_b/N_0 (dB)'); ylabel('BER');
title('BER: NRZ vs Manchester — ISI channel + LMS EQ'); legend('NRZ','Manchester');
```

OUTPUT



CONCLUSION

The MATLAB simulation demonstrated the effectiveness of different line coding schemes such as Manchester and NRZ in transmitting digital data. Manchester coding provided better synchronization and noise immunity but required more bandwidth. NRZ coding achieved higher data rates but was more susceptible to baseline drift and synchronization loss. The performance comparison confirmed that coding choice directly affects system reliability and efficiency.

PROBLEMS FACED DURING IMPLEMENTATION

Difficulty in writing MATLAB scripts to accurately simulate Manchester and NRZ waveforms. Challenges in defining appropriate sampling rates for accurate visualization. Errors in plotting and overlaying multiple signals during analysis. Confusion in interpreting the eye diagrams and BER (Bit Error Rate) outputs.

PROBLEMS RESOLVED USING SOLUTIONS

Verified each line coding waveform step-by-step using test bit sequences. Adjusted the sampling frequency and time vector for accurate waveform representation. Used MATLAB's built-in functions (stairs, plot, biterr) for accurate plotting and error analysis. Compared results with theoretical expectations to validate simulation accuracy**.**